Dr. Peet Cremer

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Senior Principal AI ~~Technical Leader with deep expertise in machine learning theory and enterprise AI strategy. Combines strong analytical~~ Researcher building time series foundation models for industrial applications. Built an AI research lab from zero, combining hands-on model development with ~~proven experience managing engineering teams (up to 15 developers)~~ research direction and ~~architecting production AI systems~~team leadership.
~~Currently leading enterprise adoption~~ Ph.D. physicist with a track record of ~~cutting-edge AI~~ turning applied ML research into production systems and advancing an AI-first, agent-enhanced development ~~tools across 100+ engineers while staying deeply hands-on in technical development~~lifecycle.

KEY RESEARCH & TECHNICAL ACHIEVEMENTS

- Led enterprise adoption of advanced Built Cognite's time series AI development tools research lab from zero and led a team of four to develop a state-of-the-art industrial time series foundation model in six months (Cursor, Gemini Code AssistApril–September 2026)across 100+ developers with ; benchmarks reported significant productivity gainsat ADIPEC 2026
- Pioneered application of computer vision algorithms to automotive perception systemsCo-authored IND-Eval, resulting in 10 patents covering training of radar perception algorithmsan industrial time series benchmark paper accepted at a NeurIPS 2026 workshop, data storage, and fleet managementa paper published at ADIPEC 2026
- Managed up to 15 developers Received Cognite's 2026 "Master of Agents" Innovation Award for leading adoption of agentic AI development tools across multiple teams while maintaining only 2 customer-impacting incidents over 1 year100+ engineers
- Architected production ML systems for building industrial knowledge graphs within an enterprise-scale data platformInventor on 5 patent applications filed with Cognite and 10 automotive patents from Aptiv; received Cognite's 2026 Inventor's Badge
- 7 peer-reviewed publications in Statistical Physics demonstrating strong research and analytical capabilities

WORK EXPERIENCE

Senior Principal Engineer – Atlas AI Researcher — COGNITE (Oslo, Norway) — 0210/2025 2026 - todaypresent

Strategic Role Transition: Moved from engineering management to Lead Cognite's industrial time series foundation model research, setting research priorities and advancing model performance on benchmarks and industrial use cases.

Principal AI Researcher — COGNITE (Oslo, Norway) — 01/2026 - 09/2026

Established the research lab, from training compute to experimental workflows. Combined hands-on technical leadership to drive AI innovation during this pivotal transformation period. Leading technical strategy for AI evaluation frameworks and enterprise AIadoption.research with leadership of a four-person team to build a state-of-the-art industrial time series foundation model in six months (April–September 2026), using an agent-enhanced software development lifecycle to accelerate iteration.

Selected contributions across both research roles (2026)

- **Synthetic Data**: Implemented synthetic time series generators based on coupled harmonic oscillators for foundation model research
- **Flow Matching**: Defined the research direction and personally designed and evaluated approaches to applying flow matching in time series foundation models
- **Evaluation**: Integrated GIFT-Eval as the primary forecasting benchmark suite; continue extending IND-Eval to evaluate performance on industrial data
- **Training Performance**: Optimized training code through architectural changes, assessed GPU options for model training, and evaluated NorMuon as an alternative to AdamW
- **Research Strategy**: Use literature review, competitor tracking, and controlled experiments to prioritize model improvements against benchmarks and industrial use cases
- **Innovation Leadership**: Serve on Cognite's Patent & Innovation Committee, vetting invention disclosures for novelty and strategic fit and contributing to patent filing decisions

SELECTED RESEARCH PUBLICATIONS

IND-Eval: Aligning Evaluation and Simulation for Industrial Time Series Foundation Models
Vebjørn Haugland, Pavel Gulin, David Foss, **Peet Cremer**, Erlend Aune, Simone Casolo, Andreas Solbrå, Milad Mobarhan. 2026.
Accepted at the Foundation Models for Temporal Systems workshop, NeurIPS 2026; presentation forthcoming, Sydney, December 2026. Industrial forecasting benchmark and study of evaluation-guided synthetic data generation.

Benchmarking Time Series Foundation Models for Oil and Gas Data Forecasting and Well Event Detection
Pavel Gulin, David Foss, Vebjørn Haugland, Erlend Aune, **Peet Cremer**, Simone Casolo.
Published in the ADIPEC 2026 proceedings; paper available on OnePetro.

Further publications in statistical physics and automotive AI: full publication list.

WORK EXPERIENCE (CONTINUED)

Principal Engineer – Atlas AI — COGNITE (Oslo, Norway) — 02/2025 - 12/2025

- **Enterprise AI StrategyAgent-Enhanced Development**: Architected organization-wide adoption of cutting-edge AI development tools (Cursor, Bolt, Led adoption of Cursor, Claude Code, Gemini Code AssistCodex, and Model Context Protocol servers), expanding usage across 100+ engineers, with teams reporting weeks-to-days productivity improvements
- **AI EvaluationInnovation**: Designing comprehensive Designed evaluation frameworks for industrial AI agents and NLP tools interfacing with Cognite's data platform
- **LLM Technical LeadershipEvaluation**: Led benchmarking of state-of-the-art models (Benchmarked models from Anthropic, OpenAI, Google, DeepSeek) and DeepSeek for industrial knowledge graph applications, directly informing Atlas AI architecture decisions . Delivered and publishing an external benchmark report providing industry insights on LLM performance for industrial AI applications
- **Cross-functional CollaborationAI Tool Governance & Rollout**: Coordinated Partnered with Legal, Security, and Procurement teams to ensure compliance, risk mitigation, and successful tool rollouts. Managed end-to-end procurement processes and conducted training sessions to maximize adoption effectiveness to introduce agentic AI tools safely and compliantly, manage costs within budget, and train teams for effective adoption

Recognized by CEO Girish Rishi in LinkedIn posts for AI tooling strategy leadership and Atlas AI engineering innovation.

Senior Director of Engineering — COGNITE (Oslo, Norway) — 02/2024 - 02/2025

Strategic engineering leadership role combining organizational growth with AI initiative development. Gained executive perspective on enterprise AI transformation while maintaining technical involvement.

- **Strategic Hiring & Expansion**: Designed and executed the hiring strategy for Cognite's India Center of ExcellenceIndia Center of Excellence, establishing staffing plans and engineering operations for 20+ roles . Contributed to successful inauguration of India Center of Excellence supporting Cognite's global expansion strategy
- **Engineering LeadershipManagement**: Managed 10–15 10–15 developers across data integration, connectivity, and AI initiatives during critical company scaling phase. Maintained only 2 customer-impacting incidents across those team's services over 1 year while scaling teams
- **Product StrategyPlanning**: Collaborated on Contributed to the Contextualization services roadmap, aligning technical architecture with business objectives

Director of Engineering — COGNITE (Oslo, Norway) — 02/2023 - 02/2024

Engineering management role focused on people leadership, technical planning, and team development. Stepped up to manage additional teams during organizational transitions when multiple Engineering Managers departed.

- **Team StabilizationLeadership**: Absorbed responsibility for multiple teams during management transitions, scaling Expanded responsibility from 1 to 3 teams (up to 18 developers) during management transitions while maintaining delivery commitments and service reliability
- **Team RecoveryDevelopment**: Transformed inherited struggling team from red-yellow Enabled support teams to yellow-green health metrics by enabling Cognite's support team to triage customer requests and creating documented guardrailed introduced documented escalation processes, significantly reducing reducing interruptions and engineering team stress
- **Technical Leadership**: Led teams specializing in contextualization of industrial data, data-driven troubleshooting applications, and parsing of engineering diagrams

Senior Machine Learning Engineer and Tech Lead — COGNITE (Oslo, Norway) — 08/2021 - 02/2023

Technical leadership role building production ML systems for building an industrial knowledge graph. Combined hands-on development with cross-functional team leadership.

- Led cross-functional team of 5 software/ML engineers implementing intelligent algorithms building production ML services for Cognite's industrial knowledge graph
- Architected and maintained microservices deploying ML algorithms in SaaS environment. Included a diagram parsingservice, a for diagram parsing, named entity recognitionservice, and a service for annotation storage , and annotation storage in a SaaS environment
- Created data infrastructure capabilities enabling advanced graph queries and industrial reality interactions Built data infrastructure supporting queries over the industrial knowledge graph

AI Lead Developer — APTIV (Wuppertal, Germany) — 12/2020 - 07/2021

Technical innovation leadership in automotive AI applications. Combined deep ML expertise with practical deployment experience.

- **Automotive AI InnovationRadar Perception**: Led breakthrough work applying Applied vision-based perception algorithms to automotive radar , using LiDAR as reference sensor. Implemented the a reference sensor; implemented point-cloud autolabeling, training dataset creationprocess, deploying a pointcloud autolabeling algorithm, and creating a robust data preprocessing and augmentation mechanism , preprocessing, and augmentation
- Led planning and execution of ML and data infrastructure projects for automotive perception tasks
- Contributed to significant innovation pipeline, resulting in 10 patents and 1 publication (see publication list)covering training of radar perception algorithmsacross the full Aptiv tenure (2017–2021), covering radar perception training, automotive data storage, and fleet management systems; also co-authored 1 publication (see publication list)

Software Development Expert — APTIV (Wuppertal, Germany) — 07/2017 - 12/2020

Built comprehensive data infrastructure and ML pipeline foundation for automotive sensor data, enabling advanced AI research and development capabilities.

- **Data Platform Engineering**: Built end-to-end data platform for automotive sensor data, including recording tools for vehicle sensors, storage infrastructure, labeling systems, and preprocessing workflows
- **Microservices for ML Inference Infrastructure**: Established microservice architecture microservices for automated inferenceprocessing, enriching incoming data with algorithm-generated insights as new sensor data is uploaded sensor data with model predictions
- **Technical Mentorship**: Supervised Master's thesis on GANs for automotive data style transfer

EDUCATION

Ph.D., Theoretical Soft Matter Physics — University of Düsseldorf — 2013 - 2017

~~**Strong Theoretical Foundation**: Deep understanding of statistical systems, numerical optimization~~Research in statistical physics, numerical simulation, and mathematical modeling~~. Research background provides analytical rigor and quantitative problem-solving skills that transfer effectively to AI /ML leadership roles~~, providing a rigorous foundation for experimental design and AI research.

- Dissertation: Mesoscale modeling of magnetic elastomers and gels using finite element methods and density functional theory
- 7 peer-reviewed publications in statistical physics (see publication list) ~~demonstrating research depth and scientific communication skills~~

M.Sc. Physics (~~GPA~~German grade: 1.1~~/4.0~~[1]) — University of Düsseldorf — 2012 - 2013

Focus: Soft Matter, Statistical Physics. Minor: Mathematics. Master~~thesis on "~~'s thesis on "Emergent states in active systems~~" published in "~~ published in a peer-reviewed journal~~, demonstrating early research impact and theoretical depth~~.

B.Sc. Physics (~~GPA~~German grade: 1.2~~/4.0~~[1]) — University of Düsseldorf — 2008 - 2012

Minor: Mathematics. ~~Strong foundation~~ Foundation in mathematical modeling and computational physics. Bachelor's thesis published as ~~journal article, showing consistent research excellence from early career~~a journal article.

Research Internship (DAAD RISE Scholarship) — Yale University, USA — 07/2010 - 10/2010

~~Selected for~~ Awarded a DAAD RISE Worldwide scholarship~~for research internship at Yale University. Conducted laboratory experiments on~~. Conducted experiments on the optical properties of butterfly wings and developed theoretical models ~~for optical behavior, demonstrating early research capabilities and international experience~~of their optical behavior.

[1] *German grading system: 1.0 (best) to 4.0 (worst)*

~~TECHNICAL~~ SKILLS & LANGUAGES

~~Python (Expert, 10+ yrs)~~ ~~C++ (Expert, 10+ yrs)~~ ~~Rust (Advanced, 2 yrs)~~ ~~TypeScript (Advanced, 2 yrs)~~

Programming Python C++ Rust TypeScript

~~PyTorch (2+ yrs)~~ ~~TensorFlow (2+ yrs)~~ ~~LangChain~~

AI Research Time Series Foundation Models Agentic AI Transformers Synthetic Data Generation Flow Matching Experimental Design Model Evaluation & Benchmarking

ML Engineering PyTorch TensorFlow scikit-learn RAG ~~Vector Databases~~ ~~Support Vector Machines~~ ~~Gradient Boosting~~ ~~Model Evaluation~~ Model Context Protocol (MCP) Production ML Systems

Scientific Computing Statistical Analysis ~~Numerical Simulations~~ Numerical Simulation Mathematical Modeling ~~NumPy/SciPy (Expert)~~ ~~Scikit-learn (2+ yrs)~~ NumPy/SciPy

~~Technical Team Leadership~~ ~~Strategic Hiring & Talent Acquisition~~ ~~Enterprise AI Strategy & Implementation~~ ~~AI Productivity Tooling Adoption~~ ~~Cross-functional Collaboration~~ ~~Operational Excellence~~ ~~Production ML Systems~~

Leadership & Management Engineering Management AI Lab Leadership Technical Hiring Technical Mentoring AI Adoption & Governance

~~Docker~~ ~~Kubernetes~~ ~~PostgreSQL~~ ~~MongoDB~~ ~~Azure~~ ~~GCP~~ ~~FastAPI~~ ~~Microservices~~ ~~Linux~~ ~~Cursor~~ ~~Claude Code~~ ~~Gemini Code Assist~~

Infrastructure & Development Docker Kubernetes Azure GCP Linux Agentic Software Development Cursor Claude Code Codex

Languages German (native) English (C1) Norwegian (B2) French (A2)

TEACHING & KNOWLEDGE SHARING

Lecturer on Artificial Intelligence

University of Wuppertal, Germany — 10/2020 - 04/2021

Designed and delivered ~~3 comprehensive~~ three lectures with exercises ~~:~~ on Numerical Optimization (~~SOTA ML optimizers like Adam & Evolutionary Algorithms~~Adam and evolutionary algorithms), Gradient Boosting, and Support Vector Machines. ~~Demonstrates ability to distill complex ML theory into teachable concepts.~~

Co-Organizer, NorwAI 2022 Hackathon

NTNU Trondheim, Norway — 08/2022 - 10/2022

Led technical organization of Data Science hackathon, including dataset curation, task definition, student supervision, and contribution evaluation.

~~TARGET ROLES & KEYWORDS~~

~~Seeking~~**Location & Availability**: ~~Staff Engineer, Principal Engineer, Technical Lead, AI Architect, Technical Director roles at technology companies with strong engineering culture. Open to hybrid remote positions in Norway or full~~ Based in Norway; open to hybrid roles in Norway and fully remote opportunities.

~~**Specializations**: AI Strategy, Machine Learning Architecture, Technical Leadership, Engineering Excellence, Enterprise AI Transformation, Industrial AI Applications, Team Scaling, Production ML Systems.~~